

Implementation of a Microservices API Gateway using Docker Compose

```
Command Prompt
#18 exporting attestation manifest sha256:f99a23cc93b11df40080eaf2f9d88aa3809e6ca7a67291472fbc96143b9684c5 0.1s done
#18 exporting manifest list sha256:c8d84277950d641a2017181baffca5272c0648fcf18787d8404d754a62654e0f 0.0s done
#18 naming to docker.io/library/api-gateway-userservice:latest
#18 naming to docker.io/library/api-gateway-userservice:latest 0.0s done
#18 unpacking to docker.io/library/api-gateway-userservice:latest 0.0s done
#18 DONE 0.3s

#19 [orderservice] exporting to image
#19 exporting layers done
#19 exporting manifest sha256:c0551984614a7bfca673ecda2c43d10dd1f5da8098526c20f90b4ce5a6560a23 done
#19 exporting config sha256:841847d97eb0ad12bb50b64557f77e138dd2fe4a9c82b84fcb412d92980df281 done
#19 exporting attestation manifest sha256:e344c02029595554633e361a60890ed8afal7b6ef5fc9f2dfa45d366f05aae87 0.1s done
#19 exporting manifest list sha256:e6e6f158c49796b0c58686de4c41a6f5168dc6b9a40df691df8459ab292d238b 0.0s done
#19 naming to docker.io/library/api-gateway-orderservice:latest done
#19 unpacking to docker.io/library/api-gateway-orderservice:latest 0.0s done
#19 DONE 0.3s

#20 [userservice] resolving provenance for metadata file
#20 DONE 0.0s

#21 [orderservice] resolving provenance for metadata file
#21 DONE 0.0s
[+] up 6/6
✓Image api-gateway-userservice Built 6.3s
✓Image api-gateway-orderservice Built 6.3s
✓Network api-gateway_default Created 0.1s
✓Container orderservice Created 0.3s
✓Container userservice Created 0.3s
✓Container gateway Created 0.3s
```

```
localhost/users/ localhost/orders/ localhost:3001 localhost:3002
localhost/users/
Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CVB in Unsupervise... Red Hat Academy System Design Primer
Pretty-print
{
  "service": "User Service",
  "message": "User service is running"
}
```

```
localhost/users/ localhost/orders/ localhost:3001 localhost:3002
localhost/orders/
Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CVB in Unsupervise... Red Hat Academy System Design Primer
Pretty-print
{
  "service": "Order Service",
  "message": "Order service is running"
}
```

```
localhost/users/ x localhost/orders/ x localhost:3001 x localhost:3002 x +
localhost:3001
{
  "service": "User Service",
  "message": "User service is running"
}
```

```
localhost/users/ x localhost/orders/ x localhost:3001 x localhost:3002 x +
localhost:3002
{
  "service": "Order Service",
  "message": "Order service is running"
}
```

```
Command Prompt
Folder PATH listing for volume New Volume
Volume serial number is 000000DF AEE6:4A19
D:.
├── docker-compose.yml
├── gateway
│   └── nginx.conf
├── orderservice
│   ├── dockerfile
│   ├── package.json
│   └── server.js
└── userservice
    ├── dockerfile
    ├── package.json
    └── server.js

D:\api-gateway>type docker-compose.yml
services:
  orderservice:
    build: ./orderservice
    container_name: orderservice
    ports:
      - "3002:3002"
    restart: unless-stopped

  userservice:
    build: ./userservice
    container_name: userservice
    ports:
      - "3001:3001"
    restart: unless-stopped

  gateway:
    image: nginx:latest
    container_name: gateway
    ports:
      - "80:80"
    restart: unless-stopped
    volumes:
      - ./gateway/nginx.conf:/etc/nginx/nginx.conf
    depends_on:
      - orderservice
      - userservice

D:\api-gateway>
```

```
Command Prompt
D:\api-gateway\orderservice>type dockerfile
FROM node:18-alpine
WORKDIR /app
COPY package.json .
RUN npm install
COPY . .
EXPOSE 3002
CMD ["node","server.js"]
D:\api-gateway\orderservice>cd ..

D:\api-gateway>cd userservice

D:\api-gateway\userservice>type dockerfile
FROM node:18-alpine
WORKDIR /app
COPY package.json .
RUN npm install
COPY . .
EXPOSE 3001
CMD ["node","server.js"]
D:\api-gateway\userservice>cd ..

D:\api-gateway>cd gatwat
The system cannot find the path specified.

D:\api-gateway>cd gateway

D:\api-gateway\gateway>type nginx.conf
events {}

http {

    server {

        listen 80;

        location /users/ {
            proxy_pass http://userservice:3001/;
        }

        location /orders/ {
            proxy_pass http://orderservice:3002/;
        }

    }
}
D:\api-gateway\gateway>
```